

REFAT KHAN

Deajeon - South Korea

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EDUCATION

Hanbat National University (HBNU)

M.Eng. in Mobile Convergence Engineering, CGPA: 4.33 / 4.5

September 2022 – September 2024

Yuseong-gu, South Korea

University of Information Technology and Science (UITS)

B.Sc. in Computer Science and Engineering, CGPA: 3.33 / 4.0

Jan. 2017 – April 2021

Dhaka, Bangladesh

RELEVANT COURSEWORK (MSC)

- Probability Theory and Random process
- Linear Algebra and Application
- Advance Digital Signals
- Convex Optimization
- Non-linear Programming
- Wireless Communications
- MIMO-OFDM
- Information Theory
- Machine Learning/Deep Learning

RELEVANT COURSEWORK (BSC)

- Machine Learning/Deep Learning
- Artificial Intelligence
- Data Structure and Algorithm
- Structured Programming Language
- Object Oriented Programming
- Operating Systems
- Computer architecture
- Database Systems Design
- Software Engineering

RESEARCH INTEREST

- Wireless Systems for Next Generation
- Precision agriculture
- Wireless Security and Privacy
- Internet of Things (IoT)
- Machine Learning for Wireless Networking
- Efficient system design for AI
- Federated Learning in Wireless Networking

SKILLS/EXPERTISE

Tools/Programming Languages: Python, C, C++, Matlab, Java, LaTeX/Overleaf, Microsoft (Word, Excel, PPT), Linux scripting, SDR (Ettus USRP) programming.

Academic/Industrial: Machine/Deep learning, analysis, research, simulations, computer networking, wireless communication networks and protocols.

Libraries/Framework: Sci-kit Learn, TensorFlow, Keras, SciPy, Pandas, NumPy, Matplotlib, Seaborn, Laravel

Interpersonal: Leadership, teaching, teamwork, active listening, responsibility, critical thinking, excellent written and oral communication skills in English Language.

RESEARCH AND PROFESSIONAL EXPERIENCE

Cognitive Communications Systems Lab

Septmber 2022 - August 2025

Graduate Student, Research Assistant

Yuseong-gu, South Korea

- Exploring vulnerabilities in the wireless network with focus on physical layer attacks and security in 5G/6G
- Reproduce results in highly ranked research papers relevant to our work, using python or matlab.
- Studying application of deep learning concepts in wireless communication.

Sabuj Lab, BRAC University

Jan. 2022 - Aug. 2022

Teaching and Research Assistant

Dhaka, Bangladesh

- Reviewed papers and conducted literature reviews.
- Organized tutorials for objective oriented Programming, Data Structure, Artificial Intelligence.
- Mentored final year undergraduate students in Machine Learning projects.
- Assisted in grading examination scripts, assignments and quizzes following guidelines given by my supervisor.

SpWave Software Firm

Sept. 2019 - Nov. 2020

Junior Web Developer

Dhaka, Bangladesh

- Assistance to construct, reconstruct, install, assemble, repair, maintain, operate or remove sub-transmission stations, electrical appliances, fittings and installations.
- Assisted in execution and supervision of national electrification programs on behalf of the Government.

Tulatuli Model High School

Jan. 2016 - Dec. 2017

Tutor

Dhaka, Bangladesh

- Taught Junior High School Students integrated science and mathematics on the average of 18 hours/week.
- Student's Welfare committee assistant head: Engaged students in talks, twice every month with regards to hygiene and cleanliness.
- Initiated a weekly science and mathematics quiz competition among students and spearheaded the activity to promote the study of the subject matter.

PROJECTS

Major Projects

Cognitive Communication System Laboratory

Dec 2022 - Jan 2025 (HBNU)

- Implement Covert Communications
- Preserve Location of important node with monitor of warden
- Exploring vulnerabilities associated with how a malicious can poison the physical layer authentication scheme during the training phase of the Machine learning-PLA

Internet-of-Things Wireless Transport Protocol

April 2023 - Oct 2023 (HBNU)

- Minimize energy consumption to prolong battery life of IoT devices
- Optimize bandwidth usage to accommodate the low data rate requirements of many IoT applications
- Provide robust encryption and authentication to safeguard data against unauthorized access and tampering

Product management at E-commerce website

Sept. 2019 - Nov. 2020 (SpWave, Dhaka)

- Data was extracted from twitter API using a library in python called tweepy and stored in a MongoDB cloud database using a library in python called pymongo.
- Libraries such as NLTK, SpellChecker and translate were used in data cleaning and preprocessing and Sklearn library was used for data analysis.
- Three classification machine learning algorithm thus K-nearest Neighbor (KNN), Support Vector Machine(SVM) and Naïve Bayes Theorem(NB) were imported from the sci-kit learn library.
- Precision, accuracy, recall, and f-1 measure were used as our evaluation models.

VOLUNTEERING EXPERIENCE

- Assisted as an international student volunteer at the ACM WiSec 2024 Conference held in South Korea
- Assisted in teaching of English Language in Korean orphanages and mentally challenged children homes
- Assisted the main teacher in providing instruction in science and Mathematics to 250 students; contributed to improving quality of instruction and student performance
- Assisted and mentored students in groups of up to 20 by reviewing their school selection and programme of study in SHS.
- Tutored students needing extra assistance for exam preparation; resulted in students improving mastery of course content

CERTIFICATIONS AND AWARDS

- The Best Paper Award (Winter conference of KICS 2024) **Feb 2024**
- Outstanding Paper Award (Radio communication conference 2023) **Sept 2023**
- Deep Learning Specialization: Coursera (taught by Prof. Andrew Ng) **Enrolled August 2024**
- Machine Learning Specialization: Coursera (taught by Prof. Andrew Ng) **April 2023**
- First Class Honors - Computer Science and Engineering (UITS). **Feb 2021**
- Most Improved Student- Computer Science and Engineering (UITS). **Feb 2021**
- Best Science and Mathematics Tutor - Tulatuli Model high school. **May 2016**

CONFERENCE PAPERS

- **Refat Khan**,Jihwan Moon, "A Channel Uncertainty-Aware Covert Transmission for a Disguised Full-Duplex Device". *Summer Annual Conference of IEIE, 2023 in Jeju Island*
- **Refat Khan**,Jihwan Moon, "Disguised Full-Duplex Transmission-Aided Covert Communications". *Radio communication conference 2023 at Hanbat National University* [**Outstanding paper award**]
- **Refat Khan**,Jihwan Moon, "Covert Communications for a Disguised Full-Duplex Vehicle With Channel Distribution Information". *KICS Winter Conference 2024 at Yong Pyong Resort, Daegwanryeong-myeon, Pyeongchang-gun, Gangwon-do* [**Best paper award**]
- **Refat Khan**,Jihwan Moon, "Transmit Power Optimization on Disguised Full-Duplex Covert Communications". *THE 34th JOINT CONFERENCE ON COMMUNICATIONS AND INFORMATION April 2024 at Busan*
- **Refat Khan**,Jihwan Moon, "Active Jamming for AF Relay-Based Covert Communications". *KIEES Radio Science and Communication Conference 2024*.

PUBLICATIONS

- **Refat Khan**, Jihwan Moon, "Detection Error Probability Maximization for Disguised Full-Duplex Covert Communications," *The Journal of Korean Institute of Communications and Information Sciences (J-KICS)*, Vol. 49, No. 12, pp. 1715–1728, Dec. 2024. DOI: 10.7840/kics.2024.49.12.1715

EXTRACURRICULAR

- Member, Youth Advocate Dhaka
- Member, Mojo 1 Fitness Center
- Member, Bangladeshi Student in South Korea Cricket Team

Language Proficiency

- Medium of Instruction for all Academic Activities is in English
- IELTS: Band Score 6.00 (expired at April)
- IELTS Preparation: Expected Band 7.5

REFERENCES

**Available on request